Songlan (Nick) Nie

Scientific Computing | Numerical PDEs | High-Performance Computing

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EDUCATION

Rice University **Aug. 2023–Present**
PhD in Computational Applied Mathematics and Operations Research; GPA: 3.95/4.0 — Advisor: Dr. Thomas Anderson
MA in Computational Applied Mathematics and Operations Research — May 2026
Thesis: Order Reduction in High-Order Integral Equation Methods for the Heat Equation

University of Oxford, Hertford College **Oct. 2019–Jun. 2023**
MMath in Mathematics and Statistics; First-Class Honours — Final-year rank: 5/37

TECHNICAL EXPERIENCE

TGS-NOPEC Geophysical Company **Houston, TX**
Returning Geophysics Research Intern **May–Aug. 2025 & May–Aug. 2026**

2026 — Total Variation Regularization for Seismic Denoising & Salt-Boundary Enhancement

- Developed a rotation-independent 3D structure-oriented directional total variation formulation to denoise reverse time migration (RTM) images while preserving sediment detail and geological reflectors.
- Implemented steerable total variation for salt-boundary enhancement in full waveform inversion (FWI); designed an in-loop regularization strategy that suppresses staircasing artifacts while preserving sediment gradients.
- Built a Fortran-to-Python interface using `f2py` for existing HPC codebases, accelerating prototyping cycles in Python.

2025 — 3D Viscoelastic Wave Modeling with Irregular Topography

- Developed a SPECFEM3D spectral element workflow for 3D viscoelastic wave modeling with irregular surface topography.
- Validated numerical accuracy against finite difference benchmarks, dispersion curve inversion, and the SEG Foothills dataset.
- Deployed GPU-accelerated SPECFEM3D workflows on enterprise HPC infrastructure, achieving $\sim 50\times$ speedup over multicore CPU baselines.

Rice University **Houston, TX**
Graduate Researcher, Computational Applied Mathematics **Aug. 2023–Present**

- Develop high-order Nyström discretizations for integral-equation formulations of PDEs in Julia using Inti.jl, implementing specialized quadrature for singular boundary and volume integral operators.
- Analyze order reduction in high-order Runge–Kutta schemes for heat and advection-diffusion equations, developing techniques that recover optimal convergence rates.

SELECTED RESEARCH PROJECTS

Newton's Method for Nonlinear Elasticity **Jun.–Aug. 2023**
Advisors: Dr. Francis Aznaran (Notre Dame), Dr. Charles Parker (Oxford)

- Derived exact Newton linearization for the St. Venant–Kirchhoff hyperelasticity model; analyzed backtracking and critical-point line search globalization strategies.
- Implemented high-order finite element discretizations in Python with Firedrake, verifying optimal theoretical convergence rates.

Finite Volume Methods for Gradient Flow PDEs **Jun.–Aug. 2022**
Advisor: Dr. Rafael Bailo (TU Eindhoven)

- Formulated an implicit–explicit (IMEX) convex-splitting finite volume scheme for L^2 gradient flow models, ensuring discrete mass conservation and energy dissipation.
- Implemented the numerical solver in Julia, reproducing expected coarsening dynamics in a crystal-growth model.

TECHNICAL SKILLS

Programming: Julia, Python, MATLAB, Fortran (`f2py`), R, SQL
Scientific Software & HPC: SPECFEM3D, Firedrake, Inti.jl, MPI, Linux clusters
Numerical Methods: Spectral element, finite element, finite volume, discontinuous Galerkin, Runge–Kutta, integral equations, optimization
Developer Tools: Git, Linux/Unix, LaTeX

LEADERSHIP & SERVICE

Rice SIAM Student Chapter — *Treasurer* **Aug. 2024–Aug. 2025**
Managed operational budgets ($1,500+) and secured annual SIAM Chapter institutional funding for academic and industry events.

SELECTED HONORS

Hertford College Scholarships (2021–2023) | Undergraduate Prize (2022) | Hertford College Summer Research Studentship (2022)